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B1.1 Cell Structures

OCR A GCSE Biology (J247) Gateway — Topic B1: Cell-level systems

1. Eukaryotic and Prokaryotic Cells

All living organisms are made of **cells**. Cells are the smallest unit of life and are classified into two fundamental types based on whether they contain a true nucleus: **eukaryotic** cells (with a membrane-bound nucleus) and **prokaryotic** cells (no true nucleus — DNA floats free in the cytoplasm).

Eukaryotic organisms include all animals, plants, fungi and protoctists. Their cells are typically 10–100 µm across and contain organelles enclosed in membranes. **Prokaryotic organisms** are bacteria and archaea — their cells are 0.1–5 µm across and contain no membrane-bound organelles.

Feature	Eukaryotic cell	Prokaryotic cell
Nucleus	Yes — membrane-bound	No — single loop of DNA in cytoplasm (nucleoid)
Other DNA	Some in mitochondria/chloroplasts	Small rings called **plasmids**
Membrane-bound organelles	Mitochondria, ER, Golgi, etc.	None
Ribosomes	Larger (80S)	Smaller (70S)
Cell wall	Plants/fungi only (cellulose/chitin)	Yes — peptidoglycan

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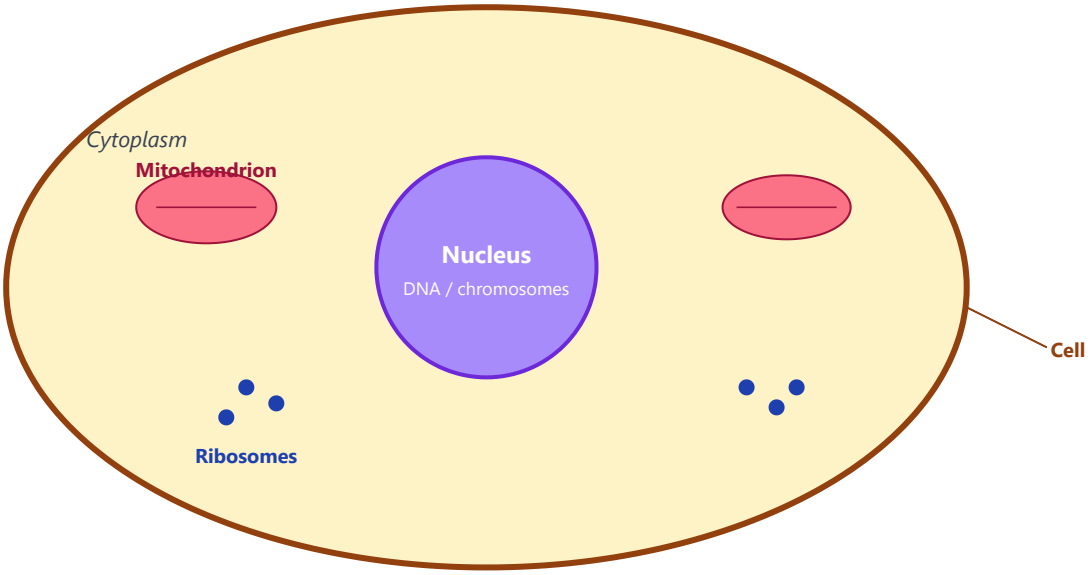
Size	10–100 µm	0.1–5 µm

2. Animal Cell — Structure and Function

Generalised Animal Cell



Organelle	Function
Nucleus	Contains the genetic material (DNA in chromosomes); controls all activities of the cell.
Cytoplasm	Jelly-like substance where most chemical reactions take place; contains dissolved nutrients, salts and enzymes.
Cell membrane	Partially permeable; controls what enters and leaves the cell.
Mitochondria	Site of **aerobic respiration** — release energy (ATP) for cell processes.
Ribosomes	Site of **protein synthesis**; assemble amino acids into proteins.

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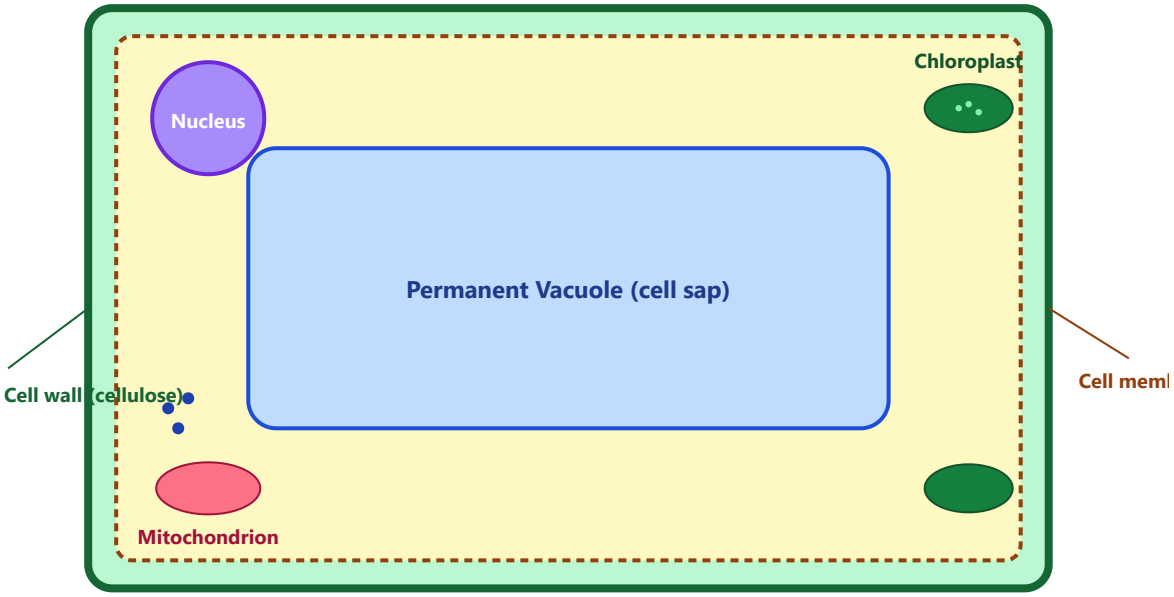
3. Plant Cell — Additional Structures

Generalised Plant Cell

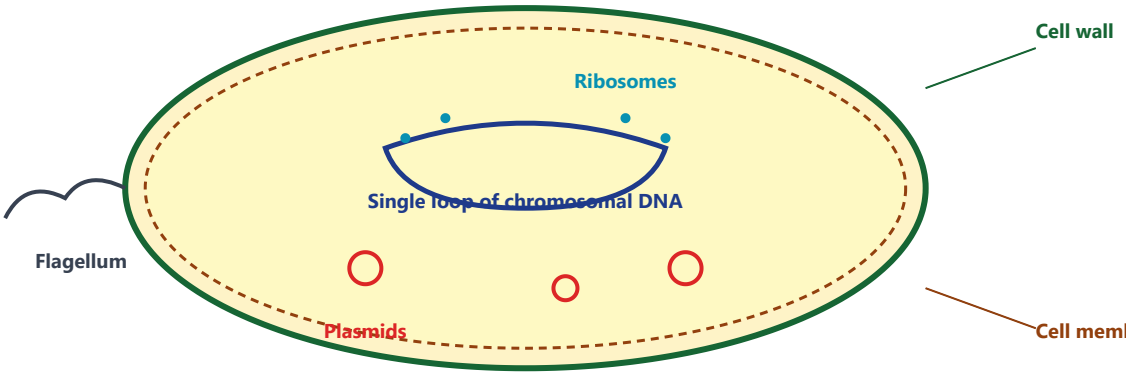


Extra structure (plant only)	Function
Cell wall (cellulose)	Strengthens and supports the cell; prevents bursting in dilute solutions.
Chloroplasts (contain chlorophyll)	Site of **photosynthesis**; absorb light energy to make glucose.
Permanent vacuole	Filled with **cell sap**; keeps the cell turgid and helps maintain the plant's shape.

Exam Tip: All plant cells also contain everything an animal cell has (nucleus, cytoplasm, cell membrane, mitochondria, ribosomes). Always state the difference: plant cells have extras — cell wall, chloroplasts, permanent vacuole. Note: not every plant cell has chloroplasts (e.g. root cells).

4. Prokaryotic (Bacterial) Cell

Bacterial (Prokaryotic) Cell



A bacterial cell has the following components: a **cell wall** (peptidoglycan, NOT cellulose), **cell membrane**, **cytoplasm**, **ribosomes**, a single loop of **chromosomal DNA**, one or more small circular **plasmids** (carry extra genes such as antibiotic resistance), and possibly a **flagellum** for movement.

5. Microscopy

Light vs Electron Microscopes

Feature	Light microscope	Electron microscope
Source	Light (visible)	Beam of electrons
Maximum magnification	×1500	×2,000,000
Resolution	~200 nm	~0.2 nm (1000× better)
Living specimens?	Yes	No — sample must be in vacuum
Cost	Cheap; portable	Very expensive; needs lab
What can be seen?	Whole cells, nucleus, chloroplasts	Sub-cellular detail: ribosomes, cristae of mitochondria, viruses

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Magnification = how much larger an object appears compared to its real size. **Resolution** = the smallest distance between two points that can still be seen as separate. Higher resolution → more detail.

$$\text{Magnification} = \frac{\text{image size}}{\text{real (actual) size}}$$

Worked example

An image of a cheek cell measures 60 mm long. The real cell is 40 µm. Find the magnification.

1. Convert to the same units: 60 mm = 60 000 µm.
2. Magnification = 60 000 ÷ 40 = **×1500**.

Exam Tip: Always check units. 1 mm = 1000 µm, 1 µm = 1000 nm. If image is in mm and real size in µm, multiply mm by 1000 first. Show your conversion in the answer for the marks.

6. Specialised Cells

Most cells in a multicellular organism are **specialised** — they have particular features that adapt them for their function. The process by which an unspecialised cell becomes specialised is called **differentiation**. Examples below.

Cell	Function	Adaptations
Sperm cell	Carry male DNA to the egg	Tail/flagellum (swims); many mitochondria (energy); acrosome with enzymes (penetrate egg); haploid nucleus (23 chromosomes)
Egg cell (ovum)	Carry female DNA; provide nutrients to embryo	Large cytoplasm with food store; haploid nucleus; membrane changes after fertilisation to block other sperm

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Ciliated epithelial cell	Move mucus along airways	Cilia (hair-like) that beat in waves; many mitochondria for energy
Red blood cell	Carry oxygen	Biconcave (large surface area); no nucleus (more room for haemoglobin); flexible to squeeze through capillaries
Nerve cell (neurone)	Carry electrical impulses	Long axon (carry signal far); branched dendrites (many connections); myelin sheath (insulation)
Muscle cell	Contract to move bones / pump blood	Many mitochondria; contractile protein fibres (actin/myosin); arranged in bundles
Root hair cell	Absorb water and mineral ions	Long projection (large surface area); thin wall (short diffusion distance); many mitochondria (active transport)
Xylem vessel	Transport water up the plant	Hollow, dead cells with no end walls (continuous tube); lignin-strengthened walls
Phloem cell (sieve tube)	Transport sugars (sucrose)	Sieve plates between cells (allow flow); companion cells supply energy

7. Required Practical Link

This topic is examined alongside **PAG B1: Microscopy**. You should be able to use a light microscope to observe and draw onion epidermal cells, then calculate the magnification of your drawing using the formula in section 5. See the separate *Required Practicals* notes for full method.

> **Exam Tip:** When asked to "describe how to use a light microscope": 1. Place slide on stage; clip in place. 2. Use lowest objective lens first. 3. Use coarse focus to bring stage close to lens, then move stage *away* while looking through eyepiece until image is clear. 4. Switch to higher magnification; use only *fine* focus. 5. Total magnification = eyepiece magnification × objective magnification.

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